

Pratyush Padhy

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EDUCATION

University of California, Irvine

Irvine, CA

B.S. Computer Science, Specialization in Intelligent Systems, GPA: 3.8/4.0

Expected June 2028

- Dean's Honor List: Fall 2025, Winter 2026, Spring 2026 | Orgs: Data@UCI, AI@UCI, Cyber@UCI, Hack@UCI

EXPERIENCE

Amazon Web Services (AWS)

Palo Alto, CA

Software Development Engineer Intern

Sept. 2026 – Present

- Designing an LLM agent to rank culprit commits for nightly Spark 4.1 regressions, graded on past regressions
- Extracting per-stage timings and flame-graph call trees from Spark event logs into JSON with Scala and Java
- Diffing each nightly run query by query to classify regressions and pinpoint the stage and operator responsible
- Surfacing each regression's diagnosis on the team's S3/CloudFront dashboard via an automated nightly trigger

Tata Consultancy Services (TCS)

Pittsburgh, PA

AI/ML Intern, Medical Robotics Center

June 2026 – Aug. 2026

- Fine-tuned SmolVLA, a vision-language-action model, to **85.3%** simulated robot pouring success over 250+ trials
- Beat a Stable-Baselines3 PPO baseline given full simulator state (**81.5%** over 3 seeds) using camera input alone
- Showed pretraining was key: the same network trained from scratch on identical input completed 0% of pours
- Ported the simulation from Gazebo to FluidLab in 3 weeks, fixing 3 engine bugs to unblock **1.5 months** of work

Ready Tutor

Irvine, CA

Software Developer Intern

March 2026 – June 2026

- Built real-time search across **1,200+** courses for Course Eater, a UCI degree planner, in React and TypeScript
- Cut database writes to **1 per action** by routing 10+ scattered storage calls through a React Context provider
- Shipped **30+** production fixes, including one for a credit-calculation bug that broke student schedule planning

PROJECTS

Chronicle | Python, FastAPI, PostgreSQL, Next.js | chronicles-weld.vercel.app

June 2026 – Sept. 2026

- Created a job aggregator ingesting **36,000+** open roles directly from 600+ Greenhouse, Lever, and Ashby boards
- Engineered hybrid full-text and pgvector search via reciprocal rank fusion at **19ms** p95 on a 20,000-job benchmark
- Raised recommendation NDCG@10 from **0.58 to 0.81** over rule-based ranking across 24 synthetic user personas
- Stripped **126MB** of raw ATS HTML and pruned stale roles to fit every role and embedding in a 512MB database

Forge | Python, PyTorch, TRL, Unsloth | forge-grpo.vercel.app

July 2026 – Sept. 2026

- Built a reinforcement learning pipeline to fine-tune Qwen2.5-1.5B-Instruct on math problems using one 8GB GPU
- Trained a LoRA adapter via GRPO, raising GSM8K pass@1 from **58.8% to 70.0%** with ARC down only 1 point
- Designed 4 verifiable rewards, fixing a zero-reward cold start with a one-shot prompt and a graded format reward
- Exported a 4-bit GGUF **3.15x** smaller than fp16 for Ollama and served base vs. tuned via vLLM multi-LoRA

FacePulse | Python, TensorFlow, OpenCV, React | face-pulse.netlify.app

April 2026 – June 2026

- Built a real-time computer vision app that classifies 5 emotions, **featured on the Handshake AI Showcase**
- Trained 4 TensorFlow models on FER2013, with a custom CNN (**64.5%**) matching a fine-tuned VGG16 (64.3%)
- Engineered dlib facial-landmark features that lifted an MLP from **43.6% to 51.7%** over raw pixel input

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, Scala, Bash, HTML/CSS

Machine Learning: PyTorch, TensorFlow, Hugging Face, vLLM, Stable-Baselines3, scikit-learn, OpenCV

Data: Apache Spark, PostgreSQL, pgvector, MongoDB, SQLite, SQLAlchemy, NumPy, Pandas, Matplotlib

Cloud & DevOps: AWS (S3, CloudFront), Google Cloud, Docker, GitHub Actions, Linux, Git

Web & Testing: FastAPI, REST APIs, React, Next.js, Tailwind CSS, pytest